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Project Size.	2.06 inch		
Model No.	PY206-W38-V2		
Display mode.	AMOLED		
Product type.	410xRGBx502 QSPI mode		
Signature by customer:			
Prepared	Checked	Approved	

Shenzhen P&O Technology Co.,Limited

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1.0 GENERAL DESCRIPTION

Item	Specification	Unit
Screen Size	2.06 inch	Diagonal
Number of Pixel	410RGB(H)x502(V)	Pixels
Display area	32.24(H)x40.48(V)	mm
Pixel pitch	0.807(H)x0.807(V)	mm
Outline Dimension	37.3x45.5x2.3±0.1	mm
Pixel arrangement	RGB Vertical Stripe	— —
Display mode	Normally Black	— —
Viewing Direction(eye)	ALL	— —
Gray inversion direction	--	
Display Color	16.7M	— —
Luminance(cd/m ²)	600	nit
Contrast Ratio	60000:1	— —
Surface treatment	— —	— —
Interface	QSPI	
Back-light	--	— —
Drive IC	CO5300AF-51	
Touch IC	CST9217	
Operation Temperature	-30~80	℃
Storage Temperature	-40~70	℃
Weight	— —	g

1.1 Features

- QSPI interface.

1.2 Applications

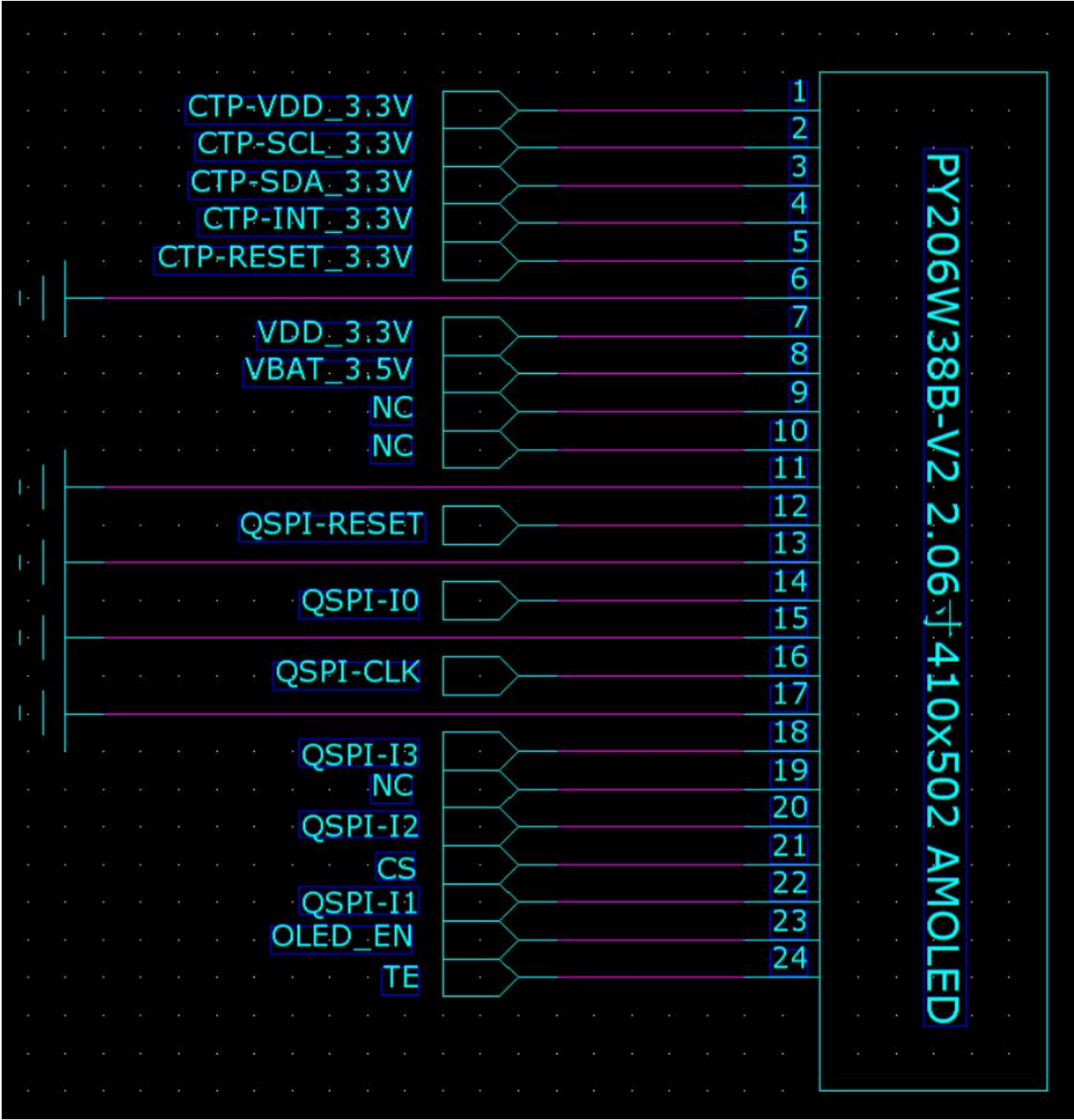
- MPOS Device.
- Personal Navigation Device.
- Other devices which require high quality displays.

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2.0 INPUT INTERFACE PIN ASSIGNMENT

FPC connector is used for electronics interface

PinNo.	Symbol	Function
1	TP-VDD 3V	Touch Power Supply
2	TP SCL 3V	Touch screen clock signal
3	TP-SDA 3V	Touch data input/output bidirectional pins
4	TP INT 3V	Touch screen interrupt signa
5	TPRST 3V	Touch screen reset signal
6	TP-GND	Ground
7	VDD 3V	Power Supply
8	VBAT	Power Supply
9	MTP/NC	NC
10	VDD 3V	Power Supply
11	GND	Ground
12	L RES	External reset input.
13	GND	Ground
14	L IOO	QSPI serial data Input
15	GND	Ground
16	L_SCLK	QSPI interface clock
17	GND	Ground
18	L IO3	QSPI serial data Input
19	NC	NC
20	L IO2	QSPI serial data Input
21	CS	Chip select input pin
22	L IO1	QSPI serial data Input
23	OLE EN	Power IC enable control pin,
24	L TE	Tearing effect output pin to synchronize MPU to frame writing



3.0 OPTICAL CHARACTERISTICS

3.1 Optical specification

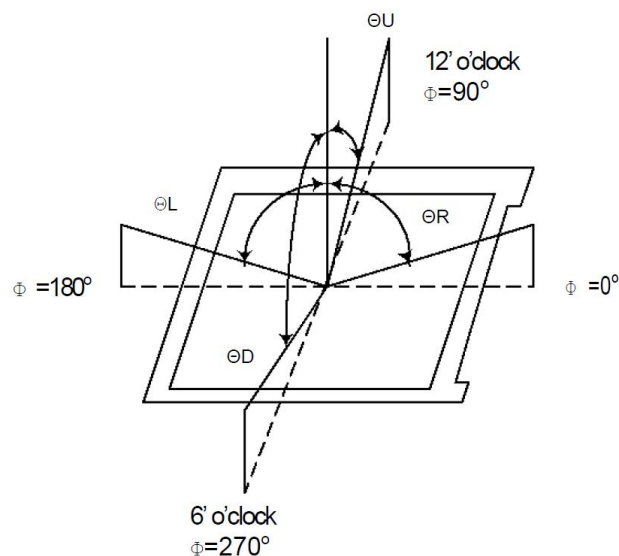
Test condition: VDDIO=1.8V , VCI=3.3 V, Ta=25℃

Item		Symbol	Condition	Value			Unit	Note
				Min	Typ	Max		
Luminance		Bp		540	600	660	nit	Note7 Normal with CG
Uniformity		Δ Bp	W255	85	-	-	%	Note 6
Viewing Angle	Left	θ_L	CR $\geq$ 10	80	85	-	Degree	Note 2
	Right	θ_R		80	85	-		
	Top	ψ_T		80	85	-		
	Bottom	ψ_B		80	85	-		
Contrast Ratio		Cr	$\Theta=0^\circ$	60000:1	-	-	-	Note 3
Color Coordinate of CIE1931 (with lens)	Red	X		0.642	0.682	0.722	-	
		Y		0.275	0.315	0.355		
	Green	X		0.190	0.240	0.290		
		Y		0.660	0.710	0.760		
	Blue	X		0.100	0.140	0.180		
		Y		0.008	0.048	0.088		

3.3 Measuring Equipment

- FPM520 of Westar Display technologies, INC., which utilized SR-3 for Chromaticity and BM-7 for other optical characteristics.
- Measuring spot size: 20 ~ 21 mm

Note (1) Definition of Viewing Angle

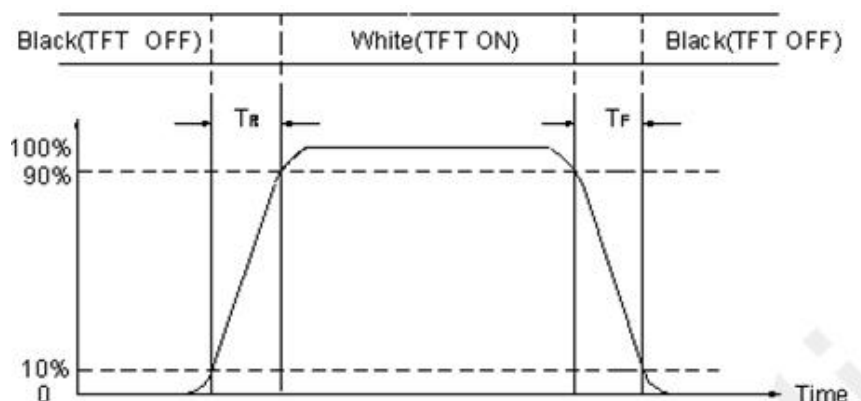


Note (2) Definition of Contrast Ratio(CR):

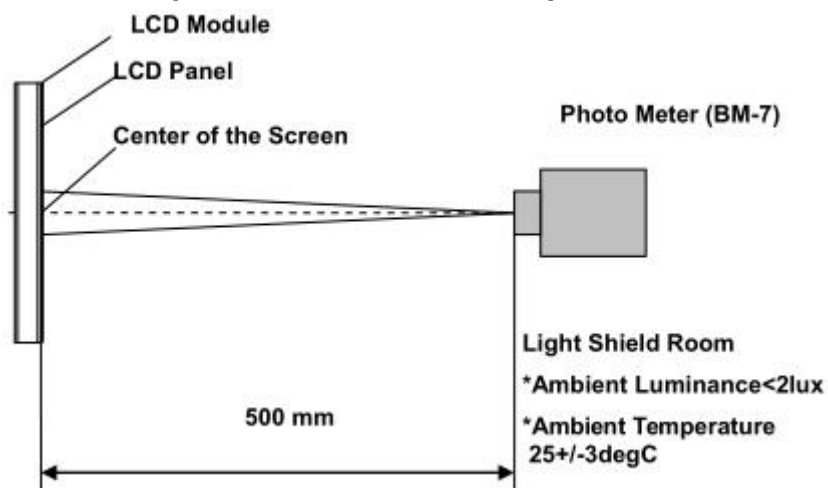
Measured at the center point of panel

$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

Note (3) Definition of Response Time: Sum of TR and TF



Note (4) Definition of optical measurement setup



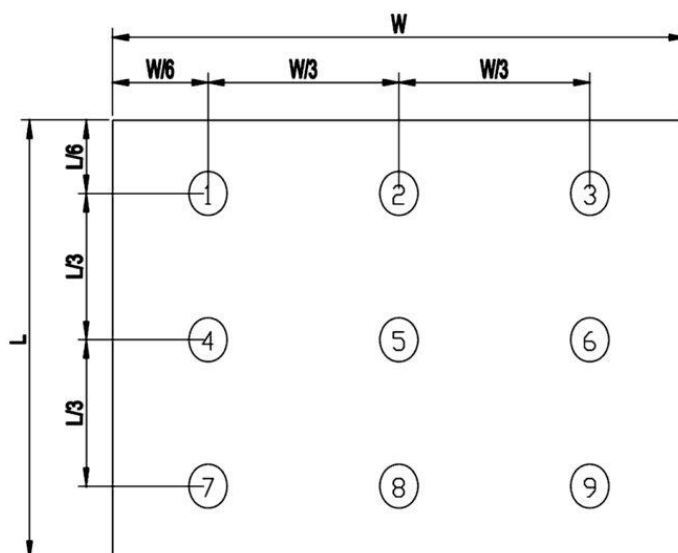
Note (5) Definition of brightness uniformity

The luminance uniformity is calculated by using following formula.

$$\Delta B_p = B_p (\text{Min.}) / B_p (\text{Max.}) \times 100 (\%)$$

$B_p (\text{Max.})$ = Maximum brightness in 9 measured spots

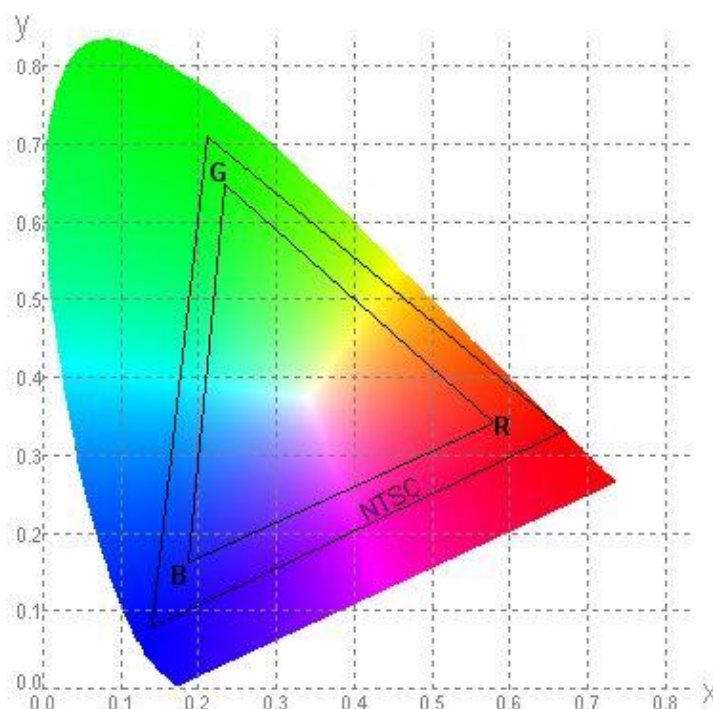
$B_p (\text{Min.})$ = Minimum brightness in 9 measured spots .



Note (6) Definition of Color of CIE1931 Coordinate and NTSC Ratio.

Color gamut:

$$S = \frac{\text{Area of RGB triangle}}{\text{Area of NTSC triangle}} \times 100\%$$



Note (7) Measured the luminance of white state at center point.

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4.0 ELECTRICAL CHARACTERISTICS

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
Analog supply voltage	VDD	2.7	3.3	3.6	V	
Digital supply voltage	VDDI	-	-	-		
Input signal Voltage	VIH	0.7VDDI	-	VDDI	V	
	VIL	GND	-	0.3VDDI	V	

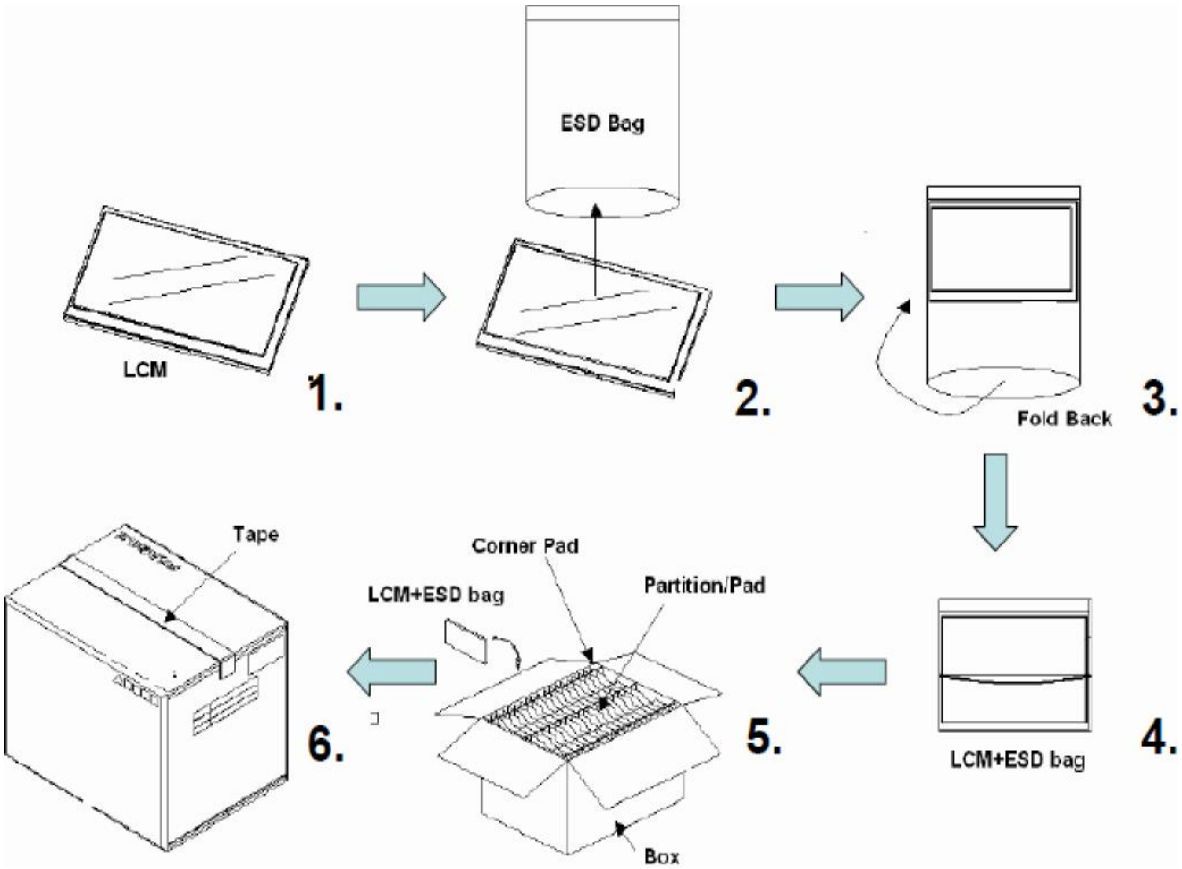
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6.0 PACKAGE SPECIFICATION

LCM Model	LCM Qty. in the box	Inner Box Size (mm)	Notice
	TDB	TDB	

6.1 Packing form

6.2 Packing assembly drawings11.2 Packing assembly drawings



Items	Material	Notice
Box	Corrugated Paper Board	AB Flute
Partition/Pad	Corrugated Paper Board	A/B Flute

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Corner Pad	Corrugated Paper Board	AB Flute
ESD bag	PE	

7.0 Items and Criteria:

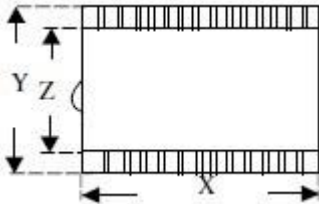
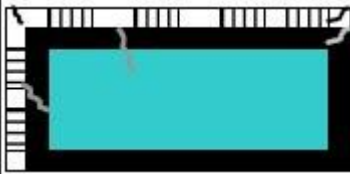
7.1 Guarantee

APEX warrants the quality of our products for **1 year** (from the date of delivery). If there are functional defects found during the period of warranty, the defective products would be replaced on a one-to-one basis. Apex would not be responsible for any direct /indirect liabilities consequential to any parties.

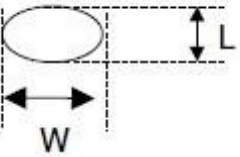
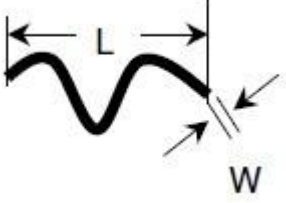
All the products should be stored or used as specified conditions described in these sheets. If module productions are not stored or used as specified conditions, herein, it will be void the **1 year** warranty(guarantee).

7.2 Visual inspection criterion in cosmetic

(1) Glass defect

Glass defect			
NO	Defect	Criteria	Remark
1	Dimension(Minor)	By engineering diagram	
2	Cracks(Major)	Extensive crack 【Reject】	

(2) LCM appearance defect

NO	Defect	Criteria		Remark
1	Round type(Minor)	Spec	Permissible Qty	1. $\psi = (L+W)/2$, L: Length, W: Width 2. Disregard if out of A.A. 
		$\psi \leq 0.10\text{mm}$	Disregard	
		$0.10\text{mm} < \psi \leq 0.20\text{mm}$	3	
		$0.20\text{mm} < \psi$	0	
2	Line type(Minor)	Spec	Permissible Qty	1. L: Length, W: Width 2. Disregard if out of A.A. 
		$W \leq 0.03\text{mm}$	Disregard	
		$L \leq 3.0\text{mm}$ and $0.03\text{mm} < W \leq 0.05\text{mm}$	2	
		$L \leq 3.0\text{mm}$ and $0.05\text{mm} < W \leq 0.10\text{mm}$	1	
		$W > 0.10\text{mm}$ or $L > 3.0\text{mm}$	0	
3		Spec.	Permissible	1. $\psi = (L+W)/2$, L: Length,

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Polarizer dent(Minor)		Qty	W: Width 2.Disregard if out of A.A.
	$\psi \leq 0.20\text{mm}$	Disregard	
	$0.20\text{mm} < \psi \leq 0.30\text{mm}$	2	
	$0.30\text{mm} < \psi \leq 0.50\text{mm}$	1	

(3) FPC

NO	Defect	Criteria	Remark
1	Copper peeling(Minor)	Copper peeling 【Reject】	
2	Golden finger	FPC golden finger broken, dead fold, indentation makes FPC surface broken 【Reject】 Tin plating layer(or gold plating) scratch, but not hurt circuit 【Accept】 Except circuit, other position scratch but not expose metal wire 【Accept】	
3	Pin	FPC PI layer delamination 【Reject】 Material and color are inconsistent with sample, FPC burrs 【Reject】 FPC Pin deformation but not affect function. 【Accept】 FPC Pin area is dirty 【Reject】 Other than FPC Pin area is dirty but not affect function 【Accept】	
4	Golden finger	Golden finger edge has burrs,foreign material 【Reject】 Golden finger oxidation (dark), uneven electroplating, pinhole, foreign material 【Reject】 Golden finger soldering pad crack exceeds 1/3 length of soldering pad, and soldering pad crack exceed 2 Pins 【Reject】 Golden finger tin plating(or gold plating)scratch, but not hurt circuit 【Accept】 Other than golden finger area scratch but not expose metal circuit 【Accept】	
5	FPC Silk printing	Ghosting, incomplete silk printing, wrong printing 【Reject】	
6	FPC Circuit line width	Line width deviation exceed 1/3 line width 【Reject】	

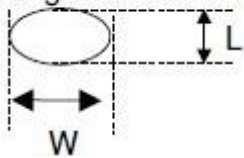
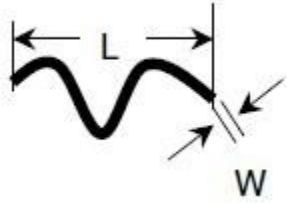
(4) Black tape

NO	Defect	Criteria	Remark
1	Shift(Minor)	IC exposed 【Reject】	
2	No black tape(Minor)	No black tape 【Reject】	

(5) Silicon

NO	Defect	Criteria	Remark
1	Amount of silicon (Minor)	ITO exposed 【Reject】	

7.3 Visual inspection criterion in electrical display

NO	Defect	Criteria		Remark
1	No display (Major)	Not allowed		
2	Missing line (Major)	Not allowed		
3	Darker or lighter Line (Major)	Not allowed		
4	Weak line(Major)	By limited sample		
5	Bright / Dark point (Minor)	Spec.	Permissible Qty	1:1sub-pixel: 1R or 1G or1B 2:Point defect area $\geq 1/2$ sub pixel.
		Bright point	1	
		Dark point	2	
6	Round type (Minor)	Spec	Permissible Qty	1. $\psi=(L+W)/2$, L: Length, W: Width 2. Disregard if out of A.A. 
		$\psi \leq 0.10\text{mm}$	Disregard	
		$0.10\text{mm} < \psi \leq 0.20\text{mm}$	3	
		$0.20\text{mm} < \psi$	0	
7	Line type (Minor)	Spec.	Permissible Qty	1. L: Length, W: Width 2. Disregard if out of A.A. 
		$W \leq 0.03\text{mm}$	Disregard	
		$L \leq 3.0\text{mm}$ and $0.03\text{mm} < W \leq 0.05\text{mm}$	2	
		$L \leq 3.0\text{mm}$ and $0.05\text{mm} < W \leq 0.10\text{mm}$	1	
		$W > 0.10\text{mm}$ or $L > 3.0\text{mm}$	0	
8	Mura (Minor)	By 5% ND filter invisible		